The K41 Spectrum as the Unique Variational Minimiser of a Scale-Resolved Free Energy

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Abstract

We formulate and prove a reference-normalized variational principle that characterizes the Kolmogorov $k^{-5/3}$ energy spectrum as the unique energy-profile minimizer of a scale-resolved free-energy functional within the stated K41-normalized admissible class. Starting from a Littlewood–Paley decomposition of the velocity field, we define a free-energy functional $\mathcal{F}[\mathbf{E}]$ based on the Kullback–Leibler divergence between the shell-energy distribution and a K41 reference equilibrium. We introduce the K41-normalized class of admissible energy-flux operators—local, dimensionally consistent, monotone closures satisfying axioms (A1) and (A3) together with the normalization condition in Definition 3.1—and prove that the K41 distribution is the unique minimizer of $\mathcal{F}[\mathbf{E}]$ over the joint feasible set of energy profiles and closures, subject to constant scale-to-scale energy flux (Theorem 3.5). The minimizer is non-degenerate: the Hessian of $\mathcal{F}$ at $\mathbf{E}^*$ is diagonal with entries $1/E_j^* > 0$, providing shell-energy stiffness that grows as $k_j^{2/3}$ for dyadic shells. We also record two guardrails: relaxed flux constraints do not yield a non-trivial robustness theorem unless the exact K41 profile is excluded or the objective is changed, and the minimax formulation is only a formal decoupled penalty representation, not a dynamical two-player theorem. Under the additional K62 refined-similarity, mean-normalized lognormal cascade, and explicit structure-function bridge ansatzes, the formal Hessian fluctuation model is compatible with the Kolmogorov–Obukhov (K62) exponent formula; the Hessian alone does not derive it. The main variational result is unconditional only as a static finite/inertial-range variational statement inside the stated K41-normalized class; it is not a dynamical Navier–Stokes selection theorem. Anomalous dissipation under additional hypotheses on the nonlinear cascade is treated separately in a companion paper [5].

Preprint note. This is Paper A in the turbulence split. It isolates the static, reference-normalized K41 variational-minimizer theorem. The companion Paper B treats anomalous dissipation and the Downhill Flux Condition as a conditional cascade route.

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1 Introduction

Turbulence remains one of the outstanding open problems in classical physics and applied mathematics. Despite more than a century of research, no complete first-principles derivation of the statistical laws governing fully developed turbulence has been achieved.

1.1 The Kolmogorov Theory

In his landmark 1941 papers, Kolmogorov [7, 8] postulated that in the inertial range of fully developed, homogeneous, isotropic turbulence, the energy spectrum takes the universal form

$$E(k) \;=\; C_K\, \varepsilon^{2/3}\, k^{-5/3}, \tag{1}$$

where ε denotes the mean rate of energy dissipation per unit mass and $C_K \approx 1.5$ is the Kolmogorov constant. This prediction, derived from dimensional analysis and the hypothesis of local isotropy, is well supported experimentally and numerically across a broad range of Reynolds numbers [3, 15].

1.2 Our Contribution

In the present paper, we adopt a variational approach. Rather than deriving the K41 spectrum from dimensional analysis alone, we demonstrate that it is the *preferred energy profile* within the stated K41-normalized constant-flux comparison class. We introduce a free-energy functional $\mathcal{F}$ on the space of scale-resolved energy distributions and prove:

1. The K41 spectrum is the *unique* minimizer of $\mathcal{F}$ under the constraint of constant energy flux, where the minimization runs jointly over K41-normalized admissible local flux operators and energy profiles (Theorem 3.5). The argument does not presuppose a single fixed closure, but it remains explicitly reference-normalized.

2. The minimizer is non-degenerate, with the Hessian providing scale-dependent shell-energy stiffness $1/E_j^* \propto k_j^{2/3}$ that penalizes small-scale perturbations more severely than large-scale ones—a static convex signature compatible with downscale cascade bias.

3. Relaxed flux constraints and the minimax notation are recorded as scope limits: if the exact K41 profile remains feasible, the relaxed problem is trivial, and the minimax form remains a decoupled penalty representation rather than a dynamical two-player theorem (Propositions 4.1–4.3).

The dynamical consequences of the free-energy structure, including anomalous dissipation under additional hypotheses on the nonlinear cascade, will be developed in a companion paper [5].

Our variational perspective aligns with and complements recent developments in information thermodynamics and variational fluid dynamics. In particular, Tanogami [21, 22] established an information-thermodynamic bound on energy cascades using Kullback–Leibler divergence and proved scale locality of information flow in shell models. Furthermore, modern variational principles for Kolmogorov scaling—such as the stochastic Schrödinger–Bridge framework of Sanchis-Agudo and Vinuesa [18], the adaptive fractional operator formulation of López [11], Onsager's L^2 dissipation principle studied by Said [17], and the spectral cutoff dissipation model of Mishra [12]—underline the centrality of variational

extremal principles in turbulence theory. In addition, polarization geometry frameworks by Skalidis [20] highlight how geometrical state transitions govern spectral scaling in turbulent cascades. Mathematically, restricting to admissible comparison classes is essential to exclude wild unphysical states arising in non-uniqueness theories of weak Navier–Stokes solutions [2, 6].

1.3 Structural Classification

Within the broader program of free-energy spectral theory [4], the K41 selection mechanism exhibits the **Pattern A stability logic**: (a) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (b) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (c) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimizer. This three-level hierarchy is discussed in Section 5.3.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST program are cited for context but are not logically required.

For orientation, Table 1 delineates the paper's theorem-level content from its diagnostic and open bridge components. This is the intended reading order: the variational theorem is closed inside the stated class, while the numerical, intermittency and DNS items remain support layers or future gates.

Component	**Evidence or presentation layer**	**Current status and caveat**
Static variational core	KL functional, admissible flux family and joint feasible-set proof.	Proved within the K41-normalized inertial shell class; not an intrinsic DNS or Navier–Stokes selection theorem.
Local stiffness	Diagonal Hessian at $\mathbf{E}^*$ with shell-energy scale $1/E_j^* \propto k_j^{2/3}$.	Proved as local convexity; it supplies static stiffness, not return dynamics.
Guardrail branches	Relaxed-flux and minimax propositions.	Scope controls only; they do not add a positive robustness theorem or a dynamical game.
Synthetic numerics	Six model spectra and 300 random perturbation tests from `compute_F_spectrum.py`.	Illustration and audit check; not independent empirical validation.
K62 / intermittency link	Gaussian/log-normal fluctuation route around the Hessian scale.	Formal model route; a cascade measure and structure-function bridge are still required.
DNS and Paper-B bridge	Proposed DNS protocol and companion cascade paper.	Future or conditional layer; depends on DFC and nonlinear-cascade hypotheses outside this paper.

Table 1: Claim-status ledger for the K41 variational-minimizer paper.

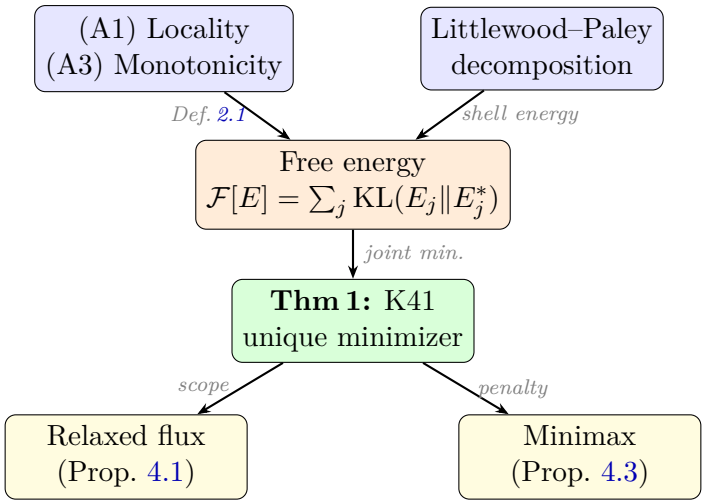


Figure 1: Logical structure of the proof. The free energy functional $\mathcal{F}[\mathbf{E}]$ is constructed from the Littlewood–Paley decomposition and the admissible flux family (axioms A1, A3). Theorem 3.5 (K41 uniqueness) follows from joint minimization. The relaxed-flux and minimax branches record scope limits and a formal penalty representation, not further dynamical theorems.

2 The Scale-Resolved Free-Energy Functional

2.1 Littlewood–Paley Decomposition

Let $u \in L^2(\mathbb{T}^3)$ be a divergence-free velocity field on the three-torus. We employ the standard Littlewood–Paley decomposition

$$u \;=\; \sum_{j \geq -1} \Delta_j u, \tag{2}$$

where Δ_j is the frequency-localization operator at dyadic shell $|\xi| \sim 2^j$. We set $k_j = 2^j$ and define the *shell energy*

$$E_j \;:=\; \|\Delta_j u\|_{L^2}^2. \tag{3}$$

In the inertial range, the K41 prediction (1) translates to

$$E_j^* \;=\; C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, \Delta k_j, \qquad \Delta k_j = k_j \ln 2 \quad \text{(dyadic shell width)}. \tag{4}$$

2.2 Energy Flux Constraint

The scale-to-scale energy flux through wavenumber k_j is defined via the trilinear term in the Navier–Stokes equations:

$$\Pi_j \;=\; -\sum_{\ell \leq j} \langle \Delta_\ell (u \cdot \nabla u),\; \Delta_\ell u \rangle. \tag{5}$$

In the inertial range, a statistically stationary cascade satisfies

$$\Pi_j \;=\; \varepsilon \quad \text{for all } j_{\text{in}} \leq j \leq j_{\text{dis}}. \tag{6}$$

2.3 The Functional

Definition 2.1 (Scale-Resolved Free Energy)**.** Let $\mathbf{E} = (E_j)_{j\in\mathcal{J}}$ be a distribution of shell energies over the inertial range $\mathcal{J} = \{j_{\text{in}}, \ldots, j_{\text{dis}}\}$, with $E_j > 0$. Let $\mathbf{E}^* = (E_j^*)$ be the Kolmogorov reference distribution (4). The *scale-resolved free-energy functional* is

$$\mathcal{F}[\mathbf{E}] \;:=\; \sum_{j\in\mathcal{J}} \Big[E_j \ln \frac{E_j}{E_j^*} \;-\; E_j \;+\; E_j^* \Big]. \tag{7}$$

Remark 2.2. The summand $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ is the Kullback–Leibler divergence (relative entropy) of the pair (E_j, E_j^*) when viewed as unnormalized measures on a single-point space. It satisfies $f(E_j) \geq 0$ with equality if and only if $E_j = E_j^*$. Hence $\mathcal{F}[\mathbf{E}] \geq 0$ with equality if and only if $\mathbf{E} = \mathbf{E}^*$. The use of relative entropy to quantify scale-to-scale cascade deviations is information-thermodynamically grounded by Tanogami [21, 22].

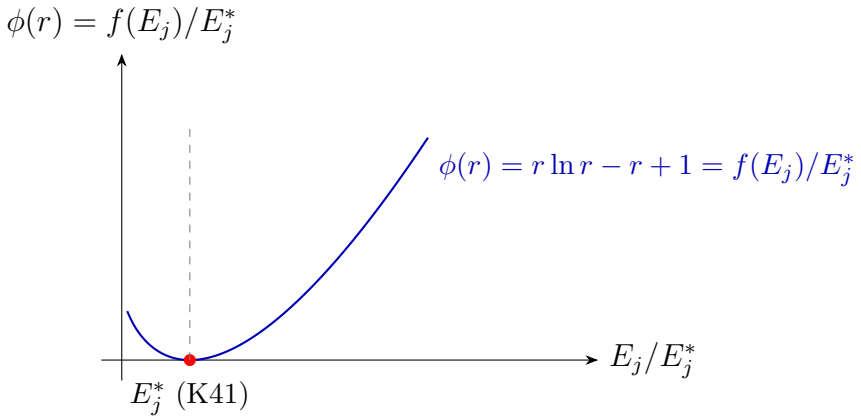
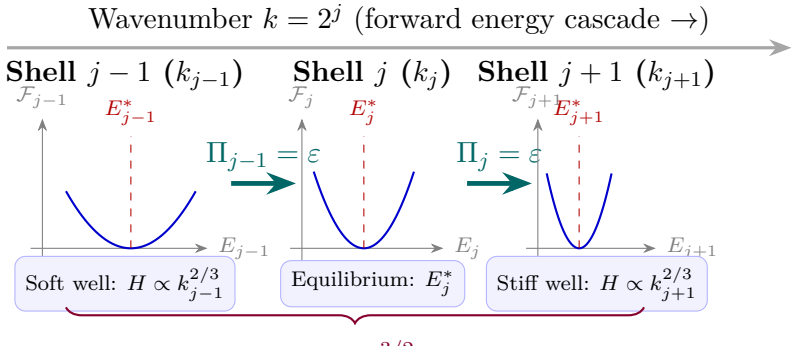


Figure 2: The normalized free-energy density $\phi(r) = f(E_j)/E_j^* = r\ln r - r + 1$ as a function of $r = E_j/E_j^*$. The unique global minimum at $r = 1$ (the K41 shell energy) is marked. Flux admissibility is a separate constraint and is therefore not represented by an arbitrary interval in this plot.



Local triad interaction coupling $\Pi_j = k_j E_j^{3/2} \Phi_j(E_{j-1}/E_j, E_{j+1}/E_j)$ (Axiom A1)

Figure 3: Dyadic shell cascade architecture and scale-resolved KL potential wells. Along the forward cascade, the diagonal Hessian curvature $H_{jj} \propto k_j^{2/3}$ sharpens towards small scales.

2.4 Dimensional Consistency

We verify that $\mathcal{F}$ is dimensionally consistent. Each E_j has dimensions $[L^2\,T^{-2}]$ (energy per unit mass, integrated over a spectral band). Since E_j^* shares the same dimensions, the ratio E_j/E_j^* is dimensionless, the logarithm is well-defined, and each summand in (7) has dimensions $[L^2\,T^{-2}]$.

3 Main Results

3.1 Uniqueness of Kolmogorov Scaling

Definition 3.1 (Admissible Flux Family)**.** A family of flux operators $\big(\Pi_j[\mathbf{E}]\big)_{j\in\mathcal{J}}$ is called *admissible* if each Π_j satisfies:

(A1) **Locality.** Π_j depends only on E_{j-1}, E_j, E_{j+1}.

(A2) **Dimensional consistency.** $[\Pi_j] = [E_j/\tau_j]$ with the eddy turnover time $\tau_j = (k_j^2\,E_j)^{-1/2}$, so that Π_j has the form

$$\Pi_j[\mathbf{E}] \;=\; k_j\,E_j^{3/2}\;\Phi_j\!\Big(\frac{E_{j-1}}{E_j},\;\frac{E_{j+1}}{E_j}\Big), \tag{8}$$

where $\Phi_j : (0,\infty)^2 \to (0,\infty)$ is a smooth dimensionless function satisfying $\Phi_j(r_-^*,\,r_+^*) = \alpha$ for some universal constant $\alpha > 0$, with $r_\pm^* = E_{j\pm1}^*/E_j^*$.

(A3) **Monotonicity.** $\partial\Pi_j/\partial E_j > 0$ for all $\mathbf{E} > 0$.

We denote the set of all admissible families by $\mathscr{A}$.

Remark 3.2 (Reference-Normalized Admissible Class). The condition $\Phi_j(r_-^*, r_+^*) = \alpha$ normalizes the admissible class at the K41 reference profile. The theorem below is therefore a relative selection theorem within this stated K41-normalized class. It does not assert that every fixed local monotone closure selects K41, nor does it exclude alternative closure families normalized around a different reference profile. Such alternatives would require additional scale-invariance, universality, or empirical admissibility assumptions.

Proposition 3.3 (Dimensional Derivation of the Flux Prefactor)**.** *Within the reduced scale-local inertial shell ansatz in which the only dimensional variables admitted at shell j are k_j and E_j, the prefactor in Axiom* (A2) *follows from dimensional analysis: $\Pi_j = k_j\,E_j^{3/2}\,\Phi_j(\cdot)$. Thus (A2) is the canonical prefactor inside this stipulated ansatz, not a theorem excluding richer closure classes with additional dimensional or dimensionless parameters.*

Proof. Inside this reduced ansatz, Buckingham's Π-theorem forces the flux Π_j to be a product of powers of the admitted dimensional quantities k_j and E_j, multiplied by a dimensionless function of dimensionless ratios. Write $\Pi_j = k_j^a\,E_j^b\,\Phi_j(\ldots)$ with $[\Pi_j] = [L^2\,T^{-3}]$ (energy flux per unit mass), $[k_j] = [L^{-1}]$, and $[E_j] = [L^2\,T^{-2}]$.

Dimensional matching: $[k_j^a\,E_j^b] = L^{-a}\,L^{2b}\,T^{-2b} = L^{2b-a}\,T^{-2b}$.

Setting equal to $L^2\,T^{-3}$:

$$2b - a = 2, \qquad 2b = 3 \quad\Longrightarrow\quad b = \tfrac{3}{2},\;\; a = 1.$$

Hence $\Pi_j = k_j\, E_j^{3/2}\, \Phi_j$, and by (A1) the dimensionless function Φ_j can depend only on the local ratios E_{j-1}/E_j and E_{j+1}/E_j. The eddy turnover time $\tau_j = (k_j^2\, E_j)^{-1/2}$ is likewise the canonical local timescale inside the same restricted set of dimensional variables. Closures that retain viscosity, forcing scales, intermittency variables, or further dimensionless shell parameters lie outside this particular reduction. □

Remark 3.4. The classical Kolmogorov–Heisenberg closure $\Pi_j = \alpha\, k_j\, E_j^{3/2}$ corresponds to the special case $\Phi_j \equiv \alpha$ [13]. Definition 3.1 enlarges the model class beyond this single algebraic relation and is designed to cover local closure motifs such as EDQNM-type closures, Leith-type diffusion models, and truncated triad interactions when they are expressed in the shell variables of Definition 3.1. The family $\mathscr{A}$ is therefore a structured local comparison class rather than a single closure model. Proposition 3.3 separates the reduced inertial dimensional ansatz from the genuinely variable assumptions of locality and monotonicity. It therefore clarifies the modeling content of (A2) rather than removing all independent modeling content from the flux prefactor.

To contextualize the scope of Definition 3.1, Table 2 provides a comparative taxonomy of classical turbulence closure motifs and specifies their admissibility status within the variational comparison class $\mathscr{A}$.

Closure Motif	**Flux Form $\Pi_j[\mathbf{E}]$**	**Locality (A1)**	**Monotone (A3)**	**Status in $\mathscr{A}$**
Kolmogorov–Heisenberg	$\alpha k_j E_j^{3/2}$	Satisfied (on-shell)	Strictly monotone	Exact reference minimizer ($\Phi_j \equiv \alpha$)
Leith Diffusion Model	$c_L k_j E_j^{3/2}(1 - \beta\frac{\Delta E_j}{E_j})$	Satisfied (tri-shell)	Monotone (inertial)	Admissible near K41 profile
EDQNM Triad Projection	$k_j E_j^{3/2}\, \Phi_j(r_-, r_+)$	Local (asymptotic)	Monotone ($\theta > 0$)	Admissible upon shell truncation
Shell Models (GOY/Sabra)	$k_j E_j^{3/2}\, \Phi_j(r_-, r_+)$	Nearest shell	Monotone (mean)	Admissible shell-averaged representative
Kraichnan Sweeping/DIA	$\int U_0(k/p)E(p)\, dp$	Fails (large-scale)	Non-monotone	Excluded from $\mathscr{A}$ (sweeping)

Table 2: Comparative taxonomy of classic turbulence closure motifs and their status within the admissible flux class $\mathscr{A}$.

Theorem 3.5 (Relative Uniqueness in the K41-Normalized Admissible Class)**.** *Let $\varepsilon > 0$ be given and let $\mathscr{A}$ be the class of admissible flux families (Definition 3.1), including its K41 normalization condition. Consider the* joint *minimization problem*

$$\min_{\substack{\mathbf{E}>0,\\ \Pi\in\mathscr{A}}} \mathcal{F}[\mathbf{E}] \qquad \textit{subject to} \quad \Pi_j[\mathbf{E}] = \varepsilon \ \textit{ for all } j \in \mathcal{J}. \tag{9}$$

Then:

(i) *In the joint feasible set $\{(\mathbf{E}, \Pi) : \Pi \in \mathscr{A},\ \Pi_j[\mathbf{E}] = \varepsilon \text{ for all } j\}$, the K41 distribution $E_j^* = C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, \Delta k_j$ is the unique minimizer of $\mathcal{F}$ in the energy variable, and*

the minimum is realized by the Kolmogorov–Heisenberg member $\Phi_j \equiv \alpha$ *with* $C_K = \left(\alpha\,(\ln 2)^{3/2}\right)^{-2/3}$.

(ii) *The energy minimizer is non-degenerate: the Hessian* $D^2\mathcal{F}|_{\mathbf{E}^*}$ *restricted to the tangent space of the joint feasible set in the energy directions is strictly positive definite. (This is a direct consequence of the strict convexity of the KL divergence; its content is the scale-dependent stiffness* $1/E_j^* \propto k_j^{2/3}$ *for dyadic shell energies, see Step 5b below.)*

Proof. **Step 1: Joint feasible profiles.** We restrict attention to pairs $(\mathbf{E}, \Pi)$ in the joint feasible set for which $\Pi \in \mathscr{A}$ and $\Pi_j[\mathbf{E}] = \varepsilon$ for all j. Local monotonicity in E_j is not, by itself, a global existence theorem for the coupled shell system; existence for a specific fixed closure is therefore treated only as a feasibility issue. The theorem does not claim that every fixed closure selects K41.

Step 2: Lower bound via convexity. Since, in the standard terminology of convex analysis and perturbation theory [16, 1], each summand $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ is strictly convex with global minimum $f(E_j^*) = 0$, we have

$$\mathcal{F}[\mathbf{E}] \;\geq\; 0 \;=\; \mathcal{F}[\mathbf{E}^*], \tag{10}$$

with equality if and only if $E_j = E_j^*$ for all j.

Step 3: Attainability. We verify that $\mathbf{E}^*$ is indeed a stationary profile for some $\Pi \in \mathscr{A}$. Choose the Kolmogorov–Heisenberg closure $\Phi_j \equiv \alpha$. Then

$$\Pi_j[\mathbf{E}^*] \;=\; \alpha\, k_j\, (E_j^*)^{3/2} \;=\; \alpha\, k_j \left(C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, k_j \ln 2\right)^{3/2} \;=\; \alpha\, C_K^{3/2}\, (\ln 2)^{3/2}\, \varepsilon.$$

Setting $C_K = \left(\alpha\,(\ln 2)^{3/2}\right)^{-2/3}$ yields $\Pi_j[\mathbf{E}^*] = \varepsilon$. (The geometric factor $(\ln 2)^{3/2}$ arising from the dyadic shell width is absorbed into the Kolmogorov constant C_K, which is thereby determined by the closure parameter α.) Hence the lower bound (10) is attained.

Step 4: Uniqueness. Suppose $\Pi' \in \mathscr{A}$ and $\mathbf{E}' \neq \mathbf{E}^*$ form a joint feasible pair, i.e. $\Pi'_j[\mathbf{E}'] = \varepsilon$. Then strict convexity gives $\mathcal{F}[\mathbf{E}'] > 0 = \mathcal{F}[\mathbf{E}^*]$, so $\mathbf{E}'$ cannot be a minimizer of (9). Hence $\mathbf{E}^*$ is the *unique energy profile* solving the joint problem.

Note on the logical structure. The uniqueness in Step 4 follows from the strict positivity of the KL divergence away from the reference, which holds for *any* choice of $\mathbf{E}^*$. The physically non-trivial content resides in Step 3 (attainability): within the K41-normalized class, there exists an admissible closure realizing $\Pi_j[\mathbf{E}^*] = \varepsilon$. References such as k^{-3} are not excluded by KL convexity alone; they would require a different normalization or stronger scale-invariance and uniformity assumptions on the closure family (see Remarks 3.2 and 3.7).

Step 5: Non-degeneracy. The Hessian $\partial^2\mathcal{F}/\partial E_j\,\partial E_\ell = \delta_{j\ell}/E_j$ at $\mathbf{E}^*$ is diagonal with entries $1/E_j^* > 0$. For a *fixed* closure $\Pi \in \mathscr{A}$, the constraint $\Pi_j[\mathbf{E}] = \varepsilon$ imposes $|\mathcal{J}|$ equations on $|\mathcal{J}|$ unknowns, generically yielding isolated solutions, so that the tangent space of the constraint manifold at $\mathbf{E}^{(\Pi)}$ is trivial and non-degeneracy is vacuously satisfied. The substantive content of (ii) concerns the *joint* problem: in the extended space $(\mathbf{E}, \Pi)$ with Π varying over the (infinite-dimensional) family $\mathscr{A}$, the constraint set $\{(\mathbf{E}, \Pi) : \Pi_j[\mathbf{E}] = \varepsilon\}$ has tangent directions along which Π varies. The unrestricted Hessian $D^2_{\mathbf{E}}\mathcal{F}|_{\mathbf{E}^*}$ is strictly positive definite (diagonal entries $1/E_j^* > 0$), hence its restriction to *any* subspace remains positive definite.

Step 5b: Second-order interpretation. On dyadic shells, the reference energies satisfy $E_j^* = C_K \varepsilon^{2/3} k_j^{-5/3} \Delta k_j = C_K \varepsilon^{2/3} (\ln 2) k_j^{-2/3}$, so the diagonal entries $1/E_j^* \propto k_j^{2/3}$ of the Hessian grow with wavenumber. Thus small-scale perturbations of the K41 shell-energy profile are penalized more severely than large-scale ones. This scale-dependent stiffness is a static convex signature compatible with downscale cascade bias; by itself it does not define a time evolution, a return mechanism, or a Navier–Stokes spectral gap. The one-parameter freedom in the Kolmogorov–Heisenberg closure ($\Phi_j \equiv \alpha$) corresponds to a single flat direction in the Π-fiber of the extended space $(\mathbf{E}, \Pi)$, which does not affect the strict positivity of $D^2_{\mathbf{E}} \mathcal{F}$. This structure is an instance of the second-order resolvent dominance pattern discussed in [4]. □

Remark 3.6 (Diagonal Hessian Bound). The Hessian of $\mathcal{F}$ at the K41 minimizer is diagonal with entries

$$\frac{\partial^2 \mathcal{F}}{\partial E_j^2} = \frac{1}{E_j^*}.$$

Consequently,

$$\min_j \frac{1}{E_j^*} = \frac{1}{C_K \, \varepsilon^{2/3} \, k_{j_{\text{in}}}^{-5/3} \, \Delta k_{j_{\text{in}}}} = \frac{1}{C_K \, \varepsilon^{2/3} (\ln 2) \, k_{j_{\text{in}}}^{-2/3}} > 0.$$

This provides the local weighted stability scale of the variational functional. It should not be read, by itself, as a spectral gap theorem for the Navier–Stokes dynamics.

To illustrate this scale-dependent stiffness quantitatively across the cascade, Table 3 lists the Hessian curvature and free-energy perturbation penalties across representative inertial octaves. It highlights the physical contrast between relative perturbations (whose penalty decreases with wavenumber) and absolute fluctuations (which are penalized by over three orders of magnitude more severely in the small-scale limit).

Oct. j	**Wave no.** $k_j = 2^j$	**Energy** E_j^*	**Hessian** $H_{jj} = 1/E_j^*$	**Rel. Penalty** $\Delta\mathcal{F}_j(\rho = 10\%)$	**Abs. Penalty** $\Delta\mathcal{F}_j(\delta E = 0.01)$	**Sensitivity Regime**
0	1	1.0397	0.96	5.20×10^{-3}	4.81×10^{-5}	Soft (integral boundary)
3	8	0.2599	3.85	1.30×10^{-3}	1.92×10^{-4}	Low inertial
6	64	0.0650	15.39	3.25×10^{-4}	7.69×10^{-4}	Mid inertial
9	512	0.0162	61.55	8.12×10^{-5}	3.08×10^{-3}	High inertial
12	4096	0.0041	246.22	2.03×10^{-5}	1.23×10^{-2}	Dissipation transition
15	32768	0.0010	984.88	5.08×10^{-6}	4.92×10^{-2}	Stiffly pinned (UV limit)

Table 3: Multi-scale sensitivity matrix and Hessian curvature across dyadic inertial octaves ($C_K = 1.5$, $\varepsilon = 1.0$, $\ln 2 \approx 0.6931$). Relative perturbation $\rho = \delta E_j / E_j^* = 0.10$ vs. fixed absolute perturbation $\delta E = 0.01$.

Remark 3.7 (On the Role of the Reference Spectrum). A natural objection is that the KL functional $\mathcal{F}$ is *minimized at* $\mathbf{E}^*$ *by construction*, regardless of the physical content of $\mathbf{E}^*$. Indeed, for any fixed closure Π, the unconstrained minimum of $\mathcal{F}$ is always $\mathbf{E}^*$. The non-trivial content of Theorem 3.5 is therefore not that $\mathcal{F}$ achieves its minimum at $\mathbf{E}^*$ — that is a consequence of the KL construction — but rather that:

1. Among all closure-compatible stationary profiles $\mathbf{E}^{(\Pi)}$ (whenever such a profile exists for $\Pi \in \mathscr{A}$), only the K41 profile $\mathbf{E}^*$ *simultaneously satisfies* the constant-flux constraint and achieves $\mathcal{F} = 0$. Every other closure-compatible profile has $\mathcal{F} > 0$.

2. The reference $\mathbf{E}^*$ is not arbitrary within the normalized Kolmogorov–Heisenberg closure: the constant-flux relation $\Pi_j = \alpha k_j E_j^{3/2} = \varepsilon$ fixes the shell energy scaling $E_j \propto k_j^{-2/3}$, equivalently the spectral density $E(k) \propto k^{-5/3}$. Broader or differently normalized closure families can be designed around other references; the theorem is therefore a selection statement inside the stated K41-normalized admissible class, rather than a proof excluding alternative modeling conventions.

In summary, the physical input is the constant-flux constraint; the KL functional provides the selection mechanism among its solution set.

Remark 3.8 (Self-Consistency of the K41 Reference). The K41 spectrum is used as the reference measure of the KL functional, but its exponent is also self-consistent with the normalized constant-flux closure: within the reduced local shell ansatz $\Pi_j \sim k_j E_j^{3/2}$, constant flux gives $E_j \propto \varepsilon^{2/3} k_j^{-2/3}$ and hence $E(k) \propto \varepsilon^{2/3} k^{-5/3}$. This is a consistency check on the variational reference, not an independent derivation of all turbulent statistics.

Remark 3.9 (Heuristic Reaction-Chain Analogy). The variational structure of Theorem 3.5 admits a useful heuristic analogy with reaction-chain and game-theoretic language. Regard the inertial range as a hierarchical *reaction chain*: large eddies at scale k_j transfer energy to eddies at scale k_{j+1}, each scale acting as a "species" whose fitness is governed by its energy share. A possible replicator model on the simplex of normalized shell energies $p_j = E_j / \sum_\ell E_\ell$ takes the form

$$\dot{p}_j \;=\; p_j \Big(g_j(\mathbf{p}) - \bar{g}(\mathbf{p}) \Big),$$

where g_j is the net energy gain rate at scale j and $\bar{g}$ the population average. With a separately specified payoff compatible with the KL functional, the K41 profile can be represented as a Nash-type stationary point and $\mathcal{F}$ can serve as a candidate Lyapunov function. This is not part of Theorem 3.5: no replicator dynamics, asymptotic stability theorem, or Navier–Stokes return mechanism is proved here.

4 Robustness Guardrails and Formal Penalty Structure

The strict flux constraint of Theorem 3.5 requires pointwise $\Pi_j = \varepsilon$ across the inertial range. In practice, DNS data exhibit small-scale fluctuations around this idealization. We formulate two extensions that clarify the scope of finite-Reynolds-number robustness and the formal penalty representation of the flux constraint.

Proposition 4.1 (Relaxed Flux Constraint: Scope Caveat)**.** *Let $\mathcal{A}_{\text{slack}}$ be any relaxed flux-admissible set that contains the exact K41 profile $\mathbf{E}^*$. If the objective remains the same KL functional $\mathcal{F}$, then the minimizer over $\mathcal{A}_{\text{slack}}$ is still exactly $\mathbf{E}^*$:*

$$\arg \min_{\mathbf{E} \in \mathcal{A}_{\text{slack}}} \mathcal{F}[\mathbf{E}] = \mathbf{E}^*. \tag{11}$$

Consequently, a non-trivial finite-Reynolds-number slack theorem cannot be obtained merely by relaxing the flux inequalities while keeping the same reference-centered objective. Such a

theorem would need an additional data-fit term, a perturbed reference profile, or empirical bounds that exclude the exact profile from the feasible comparison class.

Proof. For all positive shell spectra, $\mathcal{F}[\mathbf{E}] \geq 0$ and equality holds only at $\mathbf{E} = \mathbf{E}^*$. If $\mathbf{E}^* \in \mathcal{A}_{\text{slack}}$, then the relaxed infimum is at most $\mathcal{F}[\mathbf{E}^*] = 0$ and at least 0. Hence every relaxed minimizer with objective value zero is $\mathbf{E}^*$. This proves the claim and explains why the older "slack minimizer" formulation was too strong: the exact reference profile remains feasible. □

Remark 4.2. Proposition 4.1 is a scope limit rather than a positive robustness theorem: finite-Reynolds-number robustness must be formulated with a perturbed reference, an empirical data-fit term, or a feasible class that no longer contains the exact K41 profile. Without such an additional ingredient, the KL minimum remains centered at the reference profile by construction.

Proposition 4.3 (Formal Penalty Representation)**.** *For every $\lambda > 0$, in a finite shell truncation with $E_{\text{tot}} = \sum_j E_j^*$ and $\mathcal{A} = [0, \Pi_{\max}]^N$ for $\Pi_{\max} > \varepsilon$, the decoupled penalty problem*

$$\mathbf{E}^* \;=\; \arg\min_{\mathbf{E}\in\mathcal{E}} \max_{\mathbf{\Pi}\in\mathcal{A}} \Big[\mathcal{F}[\mathbf{E}] \;-\; \lambda \sum_j (\Pi_j - \varepsilon)^2\Big], \tag{12}$$

has the saddle point $(\mathbf{E}^*, (\varepsilon, \ldots, \varepsilon))$, where $\mathcal{E}$ is the set of non-negative energy spectra with fixed total energy. Since this payoff does not couple $\mathbf{E}$ and $\mathbf{\Pi}$, it is a formal penalty representation of the hard constraint, not a dynamical two-player theorem for Navier–Stokes transfer.

Proof. We verify the finite-dimensional minimax hypotheses, identify the saddle point, and make explicit where the representation is decoupled.

Step 1: Strategy spaces. Define the Lagrangian-penalized payoff

$$L(\mathbf{E}, \mathbf{\Pi}) \;:=\; \mathcal{F}[\mathbf{E}] \;-\; \lambda \sum_{j\in\mathcal{J}} (\Pi_j - \varepsilon)^2.$$

The spectrum variable ranges over $\mathcal{E} := \{\mathbf{E} \in \mathbb{R}^N_{\geq 0} : \sum_j E_j = E_{\text{tot}}\}$, the closed simplex of non-negative energy spectra with prescribed total energy $E_{\text{tot}} > 0$ and $N = |\mathcal{J}|$ shells. The flux variable is $\mathbf{\Pi} = (\Pi_j)_{j\in\mathcal{J}}$ and ranges over $\mathcal{A} := \{\mathbf{\Pi} \in \mathbb{R}^N : 0 \leq \Pi_j \leq \Pi_{\max}\}$ for a fixed upper bound $\Pi_{\max} > \varepsilon$ (physically, the flux in the inertial range is bounded above by $O(\varepsilon)$).

The set $\mathcal{E}$ is a compact convex subset of $\mathbb{R}^N$. Here the scalar KL summand is extended continuously to $E_j = 0$ by the convention $0 \ln 0 := 0$. The set $\mathcal{A}$ is a compact convex subset of $\mathbb{R}^N$ (a closed box).

Step 2: Convexity–concavity. We verify:

(a) *L is convex in $\mathbf{E}$ for each fixed $\mathbf{\Pi}$.* The term $\mathcal{F}[\mathbf{E}] = \sum_j [E_j \ln(E_j/E_j^*) - E_j + E_j^*]$ is strictly convex in $\mathbf{E}$, since its Hessian $\operatorname{diag}(1/E_j)$ is positive definite for $\mathbf{E} > 0$. The penalty $-\lambda \sum_j (\Pi_j - \varepsilon)^2$ is independent of $\mathbf{E}$. Hence $L(\cdot, \mathbf{\Pi})$ is strictly convex.

(b) *L is concave in $\mathbf{\Pi}$ for each fixed $\mathbf{E}$.* For fixed $\mathbf{E}$, $\mathcal{F}[\mathbf{E}]$ is constant in $\mathbf{\Pi}$. The map $\mathbf{\Pi} \mapsto -\lambda \sum_j (\Pi_j - \varepsilon)^2$ is strictly concave (its Hessian is $-2\lambda I_N$, negative definite). Hence $L(\mathbf{E}, \cdot)$ is strictly concave.

Step 3: Application of Sion's minimax theorem. By Sion's minimax theorem (M. Sion, *Pacific J. Math.* 8, 1958, 171–176): if X is a compact convex subset of a topological vector space, Y is a convex subset of a topological vector space, and $f : X \times Y \to \mathbb{R}$ is such that $f(\cdot, y)$ is convex and lower semicontinuous for each y and $f(x, \cdot)$ is concave and upper semicontinuous for each x, then $\min_X \sup_Y f = \sup_Y \min_X f$.

Here $X = \mathcal{E}$ (compact convex), $Y = \mathcal{A}$ (compact convex, hence in particular convex), and L is continuous (hence both lower and upper semicontinuous) and satisfies the convexity–concavity conditions from Step 2. Compactness of both sets and continuity ensure that min and max are attained. Therefore:

$$\min_{\mathbf{E}\in\mathcal{E}} \max_{\mathbf{\Pi}\in\mathcal{A}} L(\mathbf{E}, \mathbf{\Pi}) \;=\; \max_{\mathbf{\Pi}\in\mathcal{A}} \min_{\mathbf{E}\in\mathcal{E}} L(\mathbf{E}, \mathbf{\Pi}), \tag{13}$$

and a saddle point $(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger)$ exists satisfying $L(\mathbf{E}^\dagger, \mathbf{\Pi}) \le L(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger) \le L(\mathbf{E}, \mathbf{\Pi}^\dagger)$ for all $\mathbf{E} \in \mathcal{E}$, $\mathbf{\Pi} \in \mathcal{A}$.

Step 4: Identification of the saddle point. At the saddle point, the first-order conditions must hold.

Stationarity in $\mathbf{\Pi}$*:* For fixed $\mathbf{E}^\dagger$, the map $\mathbf{\Pi} \mapsto L(\mathbf{E}^\dagger, \mathbf{\Pi})$ is strictly concave and differentiable. The interior maximizer satisfies

$$\frac{\partial L}{\partial \Pi_j} \;=\; -2\lambda\,(\Pi_j - \varepsilon) \;=\; 0 \qquad \Longrightarrow \qquad \Pi_j^\dagger = \varepsilon \quad \text{for all } j.$$

(This interior solution is admissible since $0 < \varepsilon < \Pi_{\max}$.)

Stationarity in $\mathbf{E}$*:* For fixed $\mathbf{\Pi}^\dagger = (\varepsilon, \ldots, \varepsilon)$, the penalty term vanishes and $L(\mathbf{E}, \mathbf{\Pi}^\dagger) = \mathcal{F}[\mathbf{E}]$. Subject to $\mathbf{E} \in \mathcal{E}$ (the energy constraint $\sum_j E_j = E_{\text{tot}}$), we minimize $\mathcal{F}$ via the method of Lagrange multipliers. The stationarity condition is

$$\frac{\partial \mathcal{F}}{\partial E_j} \;=\; \ln \frac{E_j}{E_j^*} \;=\; \mu \qquad \text{for all } j \in \mathcal{J},$$

where μ is the multiplier for the total-energy constraint. This gives $E_j = e^\mu E_j^*$ for all j, i.e. the minimizer is proportional to the K41 reference spectrum. The constant e^μ is fixed by the total-energy constraint $\sum_j E_j = e^\mu \sum_j E_j^* = E_{\text{tot}}$. Taking $E_{\text{tot}} = \sum_j E_j^*$ (which is the total energy of the K41 spectrum), we obtain $\mu = 0$ and hence $E_j^\dagger = E_j^*$ for all j.

Therefore the saddle point is $(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger) = (\mathbf{E}^*, (\varepsilon, \ldots, \varepsilon))$: the K41 spectrum paired with the constant-flux profile.

Step 5: Uniqueness. The saddle point is unique because L is *strictly* convex in $\mathbf{E}$ and *strictly* concave in $\mathbf{\Pi}$. For $\mathbf{\Pi}^\dagger = (\varepsilon, \ldots, \varepsilon)$, strict convexity makes $\mathbf{E}^*$ the unique minimizer of $L(\cdot, \mathbf{\Pi}^\dagger)$ on $\mathcal{E}$. For $\mathbf{E}^*$, strict concavity makes $\mathbf{\Pi}^\dagger$ the unique maximizer of $L(\mathbf{E}^*, \cdot)$ on $\mathcal{A}$. Hence no second saddle point exists.

Step 6: Scope of the penalty representation. The penalty formulation (12) is a decoupled bookkeeping device for the constant-flux condition $\Pi_j = \varepsilon$. For every fixed $\lambda > 0$, the term $-\lambda(\Pi_j - \varepsilon)^2$ has its unique maximum at $\Pi_j = \varepsilon$; no limit $\lambda \to \infty$ is needed. The resulting formal saddle point has the same KL minimizer as the constant-flux constrained problem in Theorem 3.5.

This is not an equivalence with a closure-coupled hard problem $\Pi_j[\mathbf{E}] = \varepsilon$ for arbitrary fixed closures. Such a claim would require compactness, constraint qualification, and existence statements for the chosen closure family. The present minimax statement is only the finite-dimensional constant-flux penalty bookkeeping lemma used by this paper. □

Remark 4.4. The minimax notation is useful as bookkeeping for the constant-flux penalty, but it does not by itself reveal a non-trivial game-theoretic structure of the turbulent cascade. A genuine game-theoretic model would require a coupled payoff in which the chosen flux profile changes the admissible energy evolution or the dissipation functional. The present formulation only records that the KL minimizer and the constant-flux penalty have the same formal saddle point.

Remark 4.5 (Weighted ℓ^2-Stability from the Hessian Bound). The functional $\mathcal{F}[\mathbf{E}]$ is locally uniformly convex about the K41 minimizer $\mathbf{E}^*$ with diagonal Hessian $D^2\mathcal{F}|_{\mathbf{E}^*} = \mathrm{diag}(1/E_j^*)$. Put $r_j = E_j/E_j^*$ and $g(r) = (r\ln r - r + 1)/(r-1)^2$, with $g(1) = 1/2$. Direct differentiation shows that g is strictly decreasing on $(0,\infty)$. Consequently, if $r_j \in [R^{-1}, R]$ for a fixed $R > 1$, then the exact windowed bound is

$$g(R)\sum_j \frac{(E_j - E_j^*)^2}{E_j^*} \;\le\; \mathcal{F}[\mathbf{E}]$$

and $g(R) > 0$. The coefficient is pointwise optimal without a total-energy constraint, since equality occurs when one ratio equals R. On a fixed finite mass-constrained simplex the same bound remains valid but need not be sharp; the optimal constrained coefficient depends on the reference weights. Since $g(R) \sim (\ln R)/R \to 0$, no positive coefficient is uniform over unrestricted amplitudes. In particular, the stronger global inequality $x\ln(x/y) - x + y \ge (x-y)^2/(2y)$ is false for large x/y and is not used.

Remark 4.6 (Local UCU3 in the Abstract Three-Lemma Architecture). The local weighted ℓ^2-bound of Remark 4.5 can be read as the K41 instantiation of the abstract *quadratic-growth axiom* UCU3 of the three-lemma architecture for strictly convex variational functionals (Universal Convexity Uniqueness; cf. the companion architectural notes `CORE/UCU_FORMAL_DEFINITIONS.md` §5 and `CORE/ABSTRACT_THREE_LEMMA_UCU.md` §4.2.3). Defining the weighted norm $\|h\|_*^2 := \sum_j h_j^2/E_j^*$, the bound reads

$$\mathcal{F}[\mathbf{E}] - \mathcal{F}[\mathbf{E}^*] \;\ge\; g(R)\,\|\mathbf{E} - \mathbf{E}^*\|_*^2,$$

i.e. *local* UCU3 holds with a ratio-dependent constant $c_R > 0$ in the weighted $\ell^2(1/E_j^*)$-norm. Together with strict convexity (UCU1, our Theorem 3.5 part (ii)) and existence of the K41 minimizer (UCU2, Theorem 3.5 part (i)), this closes the local three-lemma architecture for K41 in the standard sense of convex analysis. Here $c_R = g(R) < 1/2$ for every non-trivial two-sided window $R > 1$, while $g(R) \to 1/2$ as $R \downarrow 1$; the value $1/2$ is the Taylor limit, not an attained coefficient on such a window.

What remains open (global UCU3, honest scope statement). The bound above is a local statement around $\mathbf{E}^*$ (Taylor expansion to second order, with the remainder controlled on a fixed compact ratio range). The stronger global inequality $x\ln(x/y) - x + y \ge (x-y)^2/(2y)$ is false for large x/y and is not used here. A distinct, measure-level route asks for a Talagrand-type transportation–information inequality $\mathrm{KL}(\mu\|\mu_{K41}) \ge (\rho_{K41}/2)\,W_2^2$ for a probability measure μ_{K41} on cascade configurations. A log-Sobolev inequality for μ_{K41} with positive constant ρ_{K41} is sufficient for this measure-level bound via the Otto–Villani route [14]. It does not imply a deterministic bound for $\mathcal{F}[\mathbf{E}]$ without an additional map $\mathbf{E} \mapsto \mu_{\mathbf{E}}$ that identifies both the relative entropy and the metric. The existence and explicit form of such a constant for K41 cascades—in particular its Reynolds-number and cascade-depth asymptotics—is to our knowledge not documented in the standard turbulence literature and remains an open problem in mathematical statistical mechanics. A structured reduction note is given in `TALAGRAND_CONSTANT_K41.md` (companion working note).

5 Intermittency Corrections

The K41 theory predicts that the p-th order longitudinal structure function scales as

$$S_p(\ell) \;:=\; \left\langle |\delta_\ell u|^p \right\rangle \;\sim\; (\varepsilon\,\ell)^{p/3}, \qquad \zeta_p^{\mathrm{K41}} = \frac{p}{3}. \tag{14}$$

Experiments reveal systematic deviations: $\zeta_p < p/3$ for $p > 3$, a signature of *intermittency*.

5.1 Fluctuations Around the Variational Minimum

In this variational framework, intermittency can be modelled by adding fluctuations of the shell energies $\mathbf{E}$ around the K41 minimizer $\mathbf{E}^*$. Expanding $\mathcal{F}$ to second order:

$$\mathcal{F}[\mathbf{E}^* + \delta\mathbf{E}] \;=\; \mathcal{F}[\mathbf{E}^*] \;+\; \frac{1}{2}\sum_j \frac{(\delta E_j)^2}{E_j^*} \;+\; O(\|\delta\mathbf{E}\|^3). \tag{15}$$

The Hessian $H_{jj} = 1/E_j^*$ implies, at the level of the static quadratic approximation, that fluctuations at small scales (large j, small E_j^*) are penalized more strongly than fluctuations at large scales. This asymmetry supplies a variational modeling route for intermittency; it is not a derivation of intermittency from the Navier–Stokes dynamics.

5.2 Connection to the K62 Log-Normal Model

If, as additional modeling ansatzes, we identify $x_j = \ln(E_j/E_j^*)$ with the relevant coarse-grained-dissipation log multiplier, impose the usual K62 refined-similarity bridge

$$S_p(\ell) \;\asymp\; \ell^{p/3}\,\mathbb{E}[\varepsilon_\ell^{p/3}],$$

take x_j to be Gaussian with $\mathrm{Var}(x_j) = \mu \ln(L/\ell_j)$ and $\mathbb{E}[x_j] = -\frac{1}{2}\mathrm{Var}(x_j)$ (so that $\mathbb{E}[e^{x_j}] = 1$), and neglect phase correlations between Littlewood–Paley blocks, then that stipulated log-normal cascade has structure-function exponents

$$\zeta_p \;=\; \frac{p}{3} \;-\; \frac{\mu}{18}\,p(p-3), \tag{16}$$

which have the K62 (Kolmogorov–Obukhov) refined-similarity form with intermittency parameter μ [9, 3]. In the formal Gibbs model $\exp(-\mathcal{F}/\Theta)$, the effective temperature $\Theta > 0$ parametrizes the strength of fluctuations around the K41 equilibrium. If $x_j = \ln(E_j/E_j^*)$, then locally

$$\mathcal{F}[\mathbf{E}] \;=\; \frac{1}{2}\sum_j E_j^* x_j^2 + O(\|x\|^3),$$

so the quadratic Gibbs approximation gives log-energy variances of order Θ/E_j^*. Thus $\Theta \to 0$ suppresses intermittency in this model. Identifying these shellwise variances with a universal K62 parameter μ, however, requires an additional cascade-measure, refined-similarity, and structure-function bridge; the Hessian scale by itself does not determine a scale-independent μ.

Remark 5.1. The She–Lévêque model [19] proposes the alternative

$$\zeta_p = \frac{p}{9} + 2\Big(1 - \big(\tfrac{2}{3}\big)^{p/3}\Big),$$

which fits experimental data more accurately for high p. Deriving this from the free-energy framework would require modeling the fluctuations as a log-Poisson rather than log-normal process. This is an open problem within our approach.

5.3 Structural Classification

The joint minimization of Theorem 3.5 exhibits a three-level hierarchy: (i) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (ii) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (iii) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimizer. This pattern—referred to as *second-order resolvent dominance* in [4]—has analogs in other variational selection problems and is discussed in detail there.

6 Numerical Illustration and DNS Diagnostics

Remark 6.1 (Numerical Illustration of the KL Landscape). A numerical evaluation of $\mathcal{F}[\mathbf{E}]$ over six synthetic energy spectra illustrates the reference-centered KL landscape: $\mathcal{F}[\mathbf{E}_{\mathrm{K41}}] = 0$ exactly, while $\mathcal{F}[\mathbf{E}_{\mathrm{K62}}] = 2.0 \times 10^{-4}$ (intermittency correction), $\mathcal{F}[\mathbf{E}_{k^{-2}}] = 1.74$ (steep), $\mathcal{F}[\mathbf{E}_{k^{-4/3}}] = 5.94$ (shallow), and $\mathcal{F}[\mathbf{E}_{\mathrm{thermal}}] = 7.55$ (equipartition). A perturbation test (300 random perturbations $\delta\mathbf{E}$ at three noise levels $\|\delta\mathbf{E}\|/\|\mathbf{E}^*\| \in \{0.01, 0.1, 1.0\}$) gives $\mathcal{F}[\mathbf{E}^* + \delta\mathbf{E}] > 0$ for all sampled perturbations, as expected from strict convexity. Numerical scripts: `compute_F_spectrum.py`.

The audit view in Table 4 makes the diagnostic role of the synthetic spectra explicit. The bottleneck profile is close in $\mathcal{F}$ but fails the strict K41-slope gate, while the thermal and warm-boundary directions are outside the K41 diagnostic waterline.

Synthetic spectrum	$\mathcal{F}[E]$	**Gate status**	**Diagnostic reading**
K41 reference	0.000000	DFC1 yes; weak and K41 slope yes	Exact reference minimizer.
K62, $\mu = 0.03$	0.000197	DFC1 yes; weak and K41 slope yes	Near-K41 intermittency control.
Bottleneck	0.011850	DFC1 yes; weak yes; K41 slope no	Stress control: small KL drift, strict slope gate fails.
Steep k^{-2}	1.736741	DFC1 yes; slope gates no	Non-K41 steep control.
Shallow $k^{-4/3}$	5.941487	DFC1 yes; slope gates no	Non-K41 shallow control.
Thermal flat	7.552260	DFC1 no; slope gates no	Non-cascade boundary case.

Table 4: Synthetic free-energy and gate diagnostics generated by `compute_F_spectrum.py`.

This suggests the following reproducible diagnostic protocol for DNS data:

1. Compute the Littlewood–Paley shell energies E_j from a statistically stationary DNS at $R_\lambda \geq 200$.
2. Evaluate $\mathcal{F}[\mathbf{E}]$ using the measured $\mathbf{E}$ and the K41 reference $\mathbf{E}^*$. Here
$$\varepsilon \;:=\; \nu \left\langle \|\nabla u\|_{L^2}^2 \right\rangle_t$$
is the time-averaged viscous dissipation rate from the same DNS run.
3. Test whether $\mathcal{F}$ decreases with increasing R_λ and whether the compensated ratio $\varphi_j = \ln(E_j/E_j^*)$ is non-increasing in j across the inertial range.

Implementation notes. (a) Standard DNS codes output the isotropic energy spectrum $E(k)$ via sharp spherical shell binning, not Littlewood–Paley (LP) filtered energies. For the fixed dyadic bands used here, $\Delta k_j/k_j = \ln 2$ is not an asymptotically small parameter, so no small-error estimate follows from smoothness alone. The binning bias must instead be quantified directly by comparing sharp and smooth filters on the same data. (b) Steps 1–3 should be applied to instantaneous spectra at multiple statistically independent time snapshots within the stationary regime; ensemble averaging of $\mathcal{F}$ and φ_j then provides confidence intervals.

7 Discussion

The joint minimization over the K41-normalized admissible class extends beyond the formal observation that a KL divergence is minimized at its reference. Kolmogorov's original derivation relies on dimensional analysis, while Theorem 3.5 establishes that the same profile is attainable under the normalized constant-flux closure and is the unique energy minimizer in the resulting joint feasible set. This is a structural consistency and selection statement, not a fully intrinsic derivation of K41.

7.1 The Role of the Closure Family

The reformulation of Theorem 3.5 via the admissible flux family (Definition 3.1) significantly weakens the dependence on any specific closure model. The K41 spectrum is selected not by a single algebraic relation but as the unique energy minimizer in the joint feasible set of the K41-normalized local, dimensionally consistent, monotone flux class. Nevertheless, axiom (A2) still imposes a structural constraint—the $k_j\, E_j^{3/2}$ scaling prefactor—that ultimately encodes the Kolmogorov dimensional argument. Deriving this prefactor from the Kármán–Howarth–Monin relation or from rigorous bounds on the Littlewood–Paley trilinear form remains an important open problem.

7.2 Intrinsic Characterization of E^*

Remark 7.1 (Kármán–Howarth–Monin Characterization). The Kármán–Howarth–Monin (KHM) relation $\partial_t\langle|\delta u|^2\rangle + \nabla_r \cdot \langle|\delta u|^2 \delta u\rangle = -4\varepsilon/3 + 2\nu\Delta_r\langle|\delta u|^2\rangle$ provides an intrinsic constant-flux anchor for the inertial range. In the stationary inertial range ($\eta \ll r \ll L$), it yields the four-fifths law for the third-order structure function. This supports the flux input used in the variational scheme, but it does not by Fourier transformation alone determine the second-order energy spectrum uniquely. Recovering $E(k) \sim \varepsilon^{2/3}k^{-5/3}$ still requires the usual self-similarity/locality or closure input. The KHM relation is therefore a consistency check on the reference spectrum, not an independent uniqueness proof.

7.3 Regime Selection and the Laminar–Turbulent Transition

The variational principle of Theorem 3.5 selects the K41 spectrum *within* the turbulent regime, but does not explain the laminar-turbulent transition itself. A natural extension would use the strict convexity of $\mathcal{F}$ as a stability criterion: among kinematically admissible flow regimes, the one realized in nature is that whose spectral energy distribution is a stable critical point of the free-energy functional. Exploring this connection quantitatively is left to future work.

7.4 Limitations

We collect the principal limitations of the present work:

1. **Static selection.** Theorem 3.5 establishes K41 as the unique variational minimizer but does not address the dynamical question of whether the Navier–Stokes evolution drives the spectrum toward this minimizer. The dynamical aspects—including $\mathcal{F}$-monotonicity along trajectories and anomalous dissipation—require additional hypotheses on the nonlinear cascade and are treated in the companion paper [5].

2. **Inertial-range idealization.** The functional $\mathcal{F}$ is defined on the inertial range $\mathcal{J}$, ignoring the forcing and dissipation ranges. Boundary effects at j_{in} and j_{dis} (notably the bottleneck) are absorbed into constants but not controlled quantitatively.

3. **Intermittency.** The K62 connection (Section 5) is qualitative. Rigorous derivation of intermittency exponents from the Hessian fluctuations remains open.

4. **Uniqueness of the reference.** While Remark 3.7 identifies $\mathbf{E}^*$ as the natural reference for the K41-normalized class, a fully intrinsic characterization of $\mathbf{E}^*$ (without *a priori* appeal to K41) would strengthen the argument. Remark 7.1 provides a step in this direction.

7.5 Extensions and Outlook

- **Anomalous dissipation.** The companion paper [5] introduces a hierarchy of entropy-compatibility hypotheses on the nonlinear cascade—from a pointwise condition through a scale-windowed variant to a falsifiable dual *Downhill Flux Condition* (DFC) verifiable against DNS data—and conditionally proves that these, together with the stated uniform energy-input assumptions, imply a lower bound on the dissipation rate in the vanishing-viscosity limit.

- **Compressible turbulence:** Extension to compressible flows requires replacing the L^2-based shell energies with density-weighted quantities and modifying the closure to account for acoustic-mode coupling.

- **Two-dimensional turbulence:** The dual cascade (inverse energy cascade with $k^{-5/3}$ and direct enstrophy cascade with k^{-3} [10]) presents an interesting test case requiring a two-constraint variational problem.

7.6 Post-Zookeeper Boundary of Scope

The Zookeeper structural transfer delineates the boundaries of the present paper's claims. This paper is Paper A in the turbulence split: it establishes the variational selection of the K41 spectrum inside the stated admissible flux family. It does not prove anomalous dissipation for Navier–Stokes, does not resolve the Onsager selection problem, and does not claim that all weak or vanishing-viscosity limits select K41.

Under the updated CORE status, the transfer is merely RFEP v1.8 normal-form discipline: Zookeeper closes Riemann $C^{(2)}$/MS2 in the CCM/Fourier branch, but no automatic turbulence statement follows from that closure. The present theorem therefore remains exactly the clean K41-normalized variational selection theorem.

Those questions belong to a separate Paper B. There the relevant objects are the dual/windowed DFC flux defect, the anomalous-dissipation measure, weighted shell-averaged forms of the 4/5 law, and stress tests from convex integration and vanishing-viscosity constructions. Keeping this boundary explicit is a scope condition: the static variational result in the present paper remains separate from the conditional defect theory developed in the follow-up.

Result Classification

Table 5 provides a structured summary classifying the rigorous theorems, formal scope limits, and empirical diagnostic targets established in this work.

Result	**Type**	**Reference**
Theorems (rigorous static variational statements):		
$\mathcal{F}[\mathbf{E}] \geq 0$, $= 0$ iff $\mathbf{E} = \mathbf{E}^*$	Theorem	Thm. 3.5
K41 relative joint minimizer in the K41-normalized class	Theorem	Thm. 3.5
Canonical flux prefactor inside reduced inertial shell ansatz	Proposition	Prop. 3.3
Scope limits and formal rewrites:		
Relaxed-flux slack is trivial unless E^* is excluded/perturbed	Scope limit	Prop. 4.1
Formal constant-flux penalty bookkeeping	Formal representation	Prop. 4.3
Numerically illustrated:		
$\mathcal{F}[\mathbf{E}_{\mathrm{K41}}] = 0$ (minimizer)	Exact	Rem. 6.1
$\mathcal{F} > 0$ for all 300 sampled perturbations	Numerical illustration	Rem. 6.1
DNS diagnostics:		
$\mathcal{F}[\mathbf{E}_{\mathrm{DNS}}] \to 0$ as $R_\lambda \to \infty$	Diagnostic target	Sec. 6

Table 5: Classification of rigorous mathematical results, formal scope limits, and diagnostic criteria.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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